

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Class Test

17 February 2026

Subject	Mathematics
Grade	12
Topic	Sequences and Series
Marks	50
Time	1 hour

Examiner

Mrs M Trollope

Moderator

Mrs S Rowell

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 4 questions, 4 pages and a formula sheet. Please ensure that your question paper is complete.
2. Answer ALL the questions.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers to TWO decimal places, unless stated otherwise.
8. Diagrams are NOT necessarily drawn to scale.
9. Use the mark allocation to guide the length of your solution.
10. Write neatly and legibly.
11. Rule off after each question.

QUESTION 1

A study was conducted to investigate how much time was being spent playing an online game over several days.

On Day 1, the time spent playing the game was 12 minutes.

Each day thereafter, the time spent increased by a greater amount than the previous day.

The time spent playing the game was recorded in the table below.

Day	1	2	3	4	5
Time played (minutes)	12	24	40	m	n

The sequence formed by the time spent playing the game forms a quadratic sequence.

1.1 Write down the values of m and n . (2)

1.2 Show that the time sent playing on the n^{th} day is $T_n = 2n^2 + 6n + 4$. (3)

1.3 Determine on which day the game was played for 312 minutes. (4)

[9]

QUESTION 2

The first three terms of an arithmetic series are $(2p - 3) + (p + 5) + (2p + 7)$.

The last term of the series is equal to 173.

2.1 Determine the value of p . (3)

2.2 Determine the number of terms in the series. (3)

2.3 Determine the sum of the series. (3)

2.4 If the odd numbers are removed, calculate the sum of the remaining terms in the series. (5)

[14]

QUESTION 3

3.1 Consider the geometric series: $27 + 9 + 3 + 1 + \dots$

3.1.1 Determine the general term of the series, expressing your answer as a single power of 3. (3)

3.1.2 Which term will be equal to $\frac{1}{2187}$? (3)

3.1.3 Determine the sum of the first 7 terms of the series. (2)

3.2 Consider the geometric series: $3\left(\frac{1-x}{5}\right) + 3\left(\frac{1-x}{5}\right)^2 + 3\left(\frac{1-x}{5}\right)^3 + \dots$

3.2.1 For which values of x will the series converge? (3)

3.2.2 If $x = -1$, determine the sum to infinity. (3)

[14]

QUESTION 4

4.1 The sum of the eighth term and tenth term of an arithmetic sequence is 28 and the sum of the first 12 terms is 258. Find the first term and the common difference of the sequence. (6)

4.2 Determine the value of m for which $\sum_{k=3}^m 8(2)^{k-1} > 5500$. (7)

[13]

FORMULAE

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$T_n = a + (n - 1)d$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$S_n = \frac{a(r^n - 1)}{r - 1}; r \neq 1$$

$$S_\infty = \frac{a}{1 - r}; -1 < r < 1$$

